

Aurora X200 Technical Specification

Document AX200-DS-004, Revision D, March 2026. Halcyon Dynamics product datasheet.

1. Product Overview

The Aurora X200 is an autonomous industrial inspection drone designed for unattended, dock-based operation at power substations, wind farms, ports, refineries and large construction sites. The aircraft lives in the AX200-DOCK-2 docking station, launches on a schedule or on demand, flies pre-planned inspection routes with centimetre-level positioning, and returns to the dock to recharge and upload data to the Halcyon Insight cloud platform.

The X200 combines a 45-megapixel RGB camera, a radiometric thermal imager and the AX200-LID-190 LiDAR module on a single stabilised gimbal, so a single flight can capture visual, thermal and geometric data. This document specifies the Revision D hardware shipping from March 2026 with AuroraOS firmware version 4.2.1.

2. Airframe and Dimensions

The airframe is a carbon-fibre monocoque quadrotor with folding arms and 21-inch carbon-fibre propellers (part number AX200-PROP-CF). Unfolded, the aircraft measures 1,120 x 1,120 x 380 mm including propellers. Folded for transport it measures 620 x 480 x 380 mm and fits the AX200-CASE-HD hard case.

The empty weight without batteries or payload is 6.4 kg. With two AX200-BAT-12K battery packs installed the take-off weight is 8.1 kg, and the maximum take-off weight (MTOW) is 9.2 kg. The maximum payload is therefore 2.8 kg beyond the standard sensor gimbal, which accommodates the optional gas-detection or acoustic modules.

Landing gear is fixed and includes the dock alignment pins and charging contacts. Navigation lights meet the anti-collision lighting requirement for night operations and are visible from 3 statute miles.

3. Power and Flight Performance

Power comes from two hot-swappable AX200-BAT-12K lithium-ion packs, each rated 6S 12,000 mAh (266 Wh), for a total of 532 Wh. Nominal flight time is 48 minutes with the standard gimbal and no additional payload, and 36 minutes at full 2.8 kg payload, both measured at 20 °C and sea level with a 25 percent landing reserve.

Maximum horizontal speed is 68 km/h (19 m/s) and cruise speed for inspection work is typically 5 to 8 m/s. Maximum rate of climb is 6 m/s and maximum descent rate is 4 m/s. The service ceiling is 4,000 m above mean sea level. The aircraft can hold position within 10 cm horizontally and 15 cm vertically in RTK mode.

The X200 is rated for sustained winds up to 14 m/s (50 km/h) with gusts to 18 m/s. Automatic return-to-dock is triggered when remaining battery falls to 20 percent or when measured wind exceeds the rated limit for more than 30 seconds.

4. Environmental Limits

The operating temperature range is -20 °C to +45 °C. Battery charging in the dock is permitted between -10 °C and +40 °C; outside this range the dock pre-conditions the battery compartment before charging begins. Storage temperature is -30 °C to +60 °C.

The airframe carries an IP55 ingress protection rating, allowing flight in light rain up to 10 mm per hour and dusty environments. Relative humidity from 5 to 95 percent non-condensing is supported. The gimbal optics include a heated window to prevent fogging below 5 °C.

5. Sensor Payload

The standard gimbal integrates three sensors on a 3-axis stabiliser with 0.005 degree angular vibration range:

? RGB camera: 45-megapixel full-frame sensor, 35 mm equivalent lens, mechanical shutter, 8K video at 30 frames per second, 12-bit RAW stills.

? Thermal imager (module AX200-THM-640): 640 x 512 radiometric sensor, 30 Hz, temperature measurement range -20 °C to +550 °C with accuracy of plus or minus 2 °C or 2 percent.

? LiDAR module AX200-LID-190: 190 m range at 10 percent reflectivity and 240 m at 80 percent reflectivity, 300,000 points per second, ranging accuracy of plus or minus 2 cm, 70 x 75 degree field of view.

Obstacle sensing uses six stereo camera pairs and two ultrasonic sensors to detect obstacles up to 40 m away in all directions, enabling automatic avoidance at speeds up to 12 m/s. Onboard processing is handled by a 64 TOPS AI accelerator that performs real-time anomaly detection such as hot-spot identification and corrosion flagging.

6. Connectivity and Software

The primary command-and-control (C2) link operates on 2.4 GHz and 5.8 GHz with AES-256 encryption and a line-of-sight range of 8 km. A built-in 5G/LTE modem with dual SIM provides beyond-line-of-sight telemetry and video streaming, and Wi-Fi 6 is used for high-speed data offload while docked. Wired Gigabit Ethernet is available on the docking station.

Positioning uses multi-constellation RTK GNSS (GPS, Galileo, GLONASS and BeiDou) with 1 cm + 1 ppm horizontal accuracy when a correction source is available, backed by visual-inertial odometry for GNSS-denied environments such as under bridges.

The aircraft ships with AuroraOS firmware version 4.2.1 and the dock with DockOS 2.6.0. Mission planning, live monitoring and data review are performed in the Halcyon Insight cloud platform or through the REST and MQTT APIs. Firmware updates are delivered over the air and are signed; the aircraft refuses to boot unsigned images (see error code E-999 in the Installation and Maintenance Guide).

7. Docking Station AX200-DOCK-2

The AX200-DOCK-2 docking station is a weatherproof, climate-controlled enclosure that charges, protects and launches the aircraft. It measures 1,300 x 1,100 x 700 mm with the lid closed, weighs 96 kg and requires a single-phase 240 V AC supply at 50 or 60 Hz on a dedicated 16 A circuit, drawing up to 1.8 kW at peak.

Charging time from empty to 100 percent is 55 minutes; an 80 percent charge takes 35 minutes. The dock includes an integrated weather station, an RTK base station, a backup battery that closes the lid during a power failure, and a 5G modem. Mounting requires four M12 anchor bolts torqued to 65 Nm as detailed in the Installation and Maintenance Guide.

8. Safety and Certifications

The Aurora X200 carries CE marking, complies with FCC Part 15 Subpart B, and is certified as a class C3 unmanned aircraft under EN 4709-001. The design conforms to ISO 21384-3 operational procedures and the battery packs are certified to IEC 62133-2 and UN 38.3 for transport.

Safety features include geofencing with configurable altitude and boundary limits, dual redundant IMUs and flight controllers, automatic return-to-dock at 20 percent battery, remote ID broadcast, and the optional AX200-PARA-01 ballistic parachute, which deploys in 0.8 seconds and limits descent to 5 m/s. The aircraft produces 72 dB(A) of noise measured at 3 m in hover.

9. Operating Modes and Mission Planning

The Aurora X200 supports three operating modes. Scheduled mode flies a stored inspection route at configured times, up to 12 missions per day per dock. On-demand mode launches within 90 seconds of a request from Halcyon Insight or the API, for example when a SCADA alarm is raised. Patrol mode flies a continuous perimeter route and returns automatically when the battery reaches 20 percent.

A mission may contain up to 500 waypoints and 2,000 capture actions, and a single dock stores up to 200 missions. Missions are planned in 3D against the customer's site model with automatic terrain following at a configurable height above ground of 5 to 120 m. The aircraft requires at least 8 satellites and an RTK fix before every launch, and the dock aborts a launch when wind measured at the dock exceeds 12 m/s.

10. Data Handling and Storage

The aircraft carries a 1 TB solid-state drive encrypted with AES-256; a full 48-minute mission generates roughly 60 GB of imagery and point-cloud data. Data is transferred to the dock over Wi-Fi 6 at up to 1.2 Gbit/s after landing and uploaded to Halcyon Insight, where it is retained for 24 months by default. Customers may instead deploy the Halcyon Insight Edge appliance to keep all data on their own premises.

All communication with Halcyon Insight uses TLS 1.3. Flight logs for the last 200 flights are retained on the aircraft and are required for warranty claims. The Remote ID broadcast contains the aircraft serial number, position and take-off location as required by regulation.

11. Specification Summary

Model: Aurora X200, Revision D.

Airframe type: Carbon-fibre quadrotor with folding arms.

Dimensions unfolded: 1,120 x 1,120 x 380 mm.

Dimensions folded: 620 x 480 x 380 mm.

Empty weight: 6.4 kg without batteries.

Maximum take-off weight: 9.2 kg.

Maximum payload: 2.8 kg.

Battery: 2 x AX200-BAT-12K, 6S 12,000 mAh, 532 Wh total.

Flight time: 48 minutes without payload, 36 minutes at full payload.

Maximum speed: 68 km/h.

Maximum wind resistance: 14 m/s sustained, 18 m/s gusts.

Service ceiling: 4,000 m above mean sea level.

Operating temperature: -20 °C to +45 °C.

Ingress protection: IP55.

RGB camera: 45 megapixel full-frame, 8K video.

Thermal imager: 640 x 512 radiometric, module AX200-THM-640.

LiDAR: 190 m range at 10 percent reflectivity, module AX200-LID-190.

Obstacle sensing: Stereo and ultrasonic, 40 m range.

Positioning: RTK GNSS, 1 cm + 1 ppm horizontal.

C2 link: 2.4 and 5.8 GHz, AES-256, 8 km line of sight.

Cellular: 5G/LTE modem with dual SIM.

Firmware: AuroraOS 4.2.1 aircraft, DockOS 2.6.0 dock.

Charging time: 55 minutes to 100 percent, 35 minutes to 80 percent.

Dock power: 240 V AC, 50/60 Hz, 1.8 kW peak.

Dock dimensions and weight: 1,300 x 1,100 x 700 mm, 96 kg.

Noise: 72 dB(A) at 3 m in hover.

Certifications: CE, FCC Part 15 Subpart B, EN 4709-001 class C3.

Standard warranty: 24 months from delivery.

12. Ordering Information and Part Numbers

The following part numbers are used for ordering and for warranty claims. Serial numbers are printed on the label inside the battery bay and are also readable from the Halcyon Insight device page.

AX200-BASE-01: Aurora X200 airframe with standard sensor gimbal.

AX200-BAT-12K: Battery pack, 6S 12,000 mAh, 266 Wh.

AX200-DOCK-2: Docking station with RTK base and weather station.

AX200-PROP-CF: Set of four 21-inch carbon-fibre propellers.

AX200-LID-190: LiDAR module, 190 m range.

AX200-THM-640: Radiometric thermal imager module.

AX200-PARA-01: Ballistic parachute recovery system.

AX200-CASE-HD: Hard transport case for folded aircraft.

A standard Aurora X200 system consists of one AX200-BASE-01, two AX200-BAT-12K packs, one AX200-DOCK-2 and one spare AX200-PROP-CF set. Systems are shipped from Meridian Falls, Colorado with a typical lead time of 8 weeks.

13. Warranty Summary

Every Aurora X200 system is covered by a standard warranty of 24 months from the date of delivery covering defects in materials and workmanship. The optional AuroraCare Plus service plan extends coverage to 36 months and includes one preventive service visit per year. Battery packs are covered for 12 months or 300 charge cycles, whichever comes first.

Full warranty terms, exclusions and the claims procedure are published in section 13 of the Aurora X200 Installation and Maintenance Guide (document AX200-IM-003). Specifications in this datasheet are subject to change without notice; the Halcyon Insight documentation portal always carries the current revision.